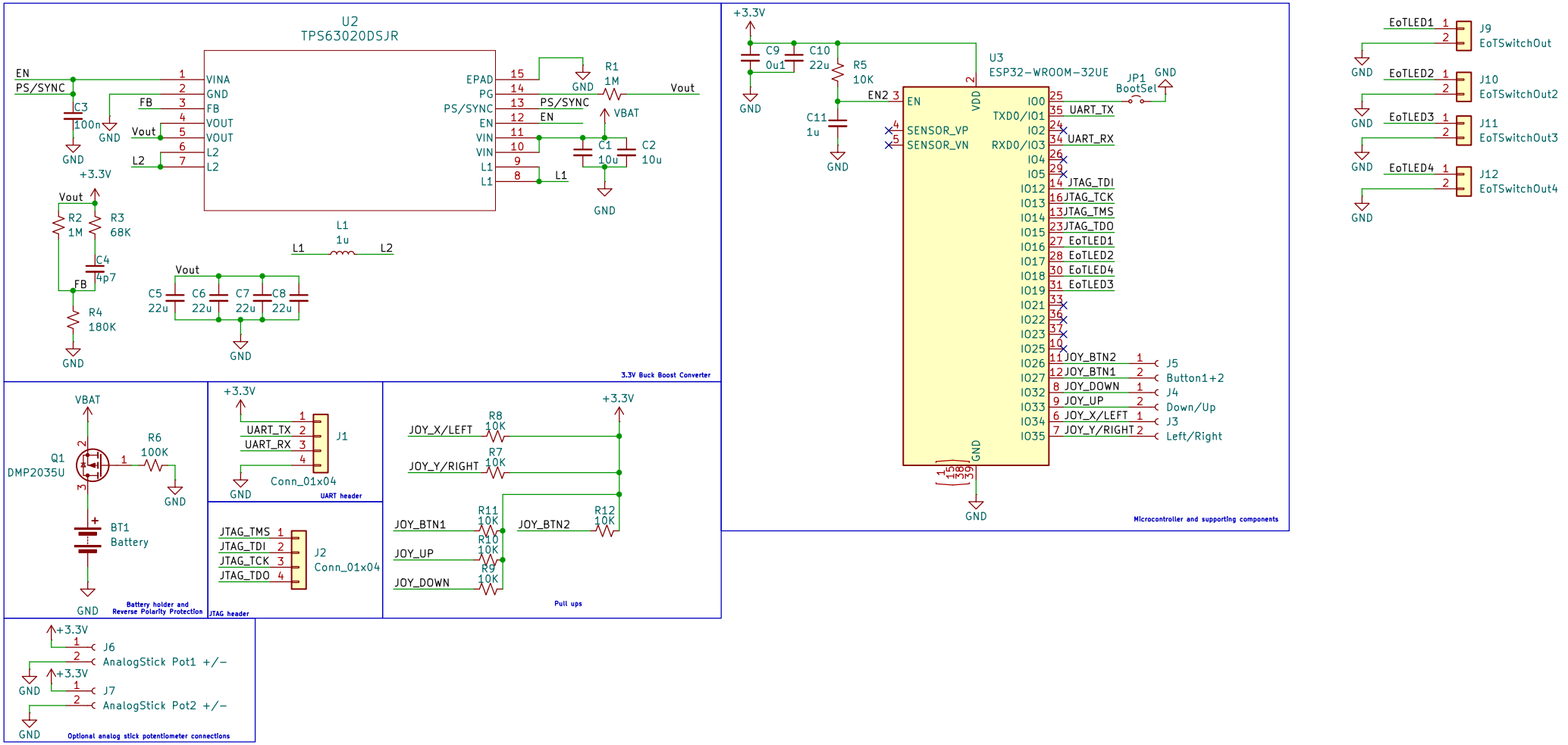
U2
TPS63020DSJR
EN
PS/SYNC
FB
Vout
L2
VINA
GND
FB
VOUT
VOUT
L2
L2
EPAD
PG
PS/SYNC
EN
VIN
VIN
L1
L1
GND
PS/SYNC
EN
R1
1M
VBAT
C1
10u
C2
10u
C3
100n
R2
1M
R3
68K
C4
4p7
R4
180K
L1
1u
C5
22u
C6
22u
C7
22u
C8
22u
+3.3V
3.3V Buck Boost Converter
U3
ESP32-WROOM-32UE
C9
0u1
C10
22u
C11
1u
R5
10K
EN2
JP1
BootSel
SENSOR_VP
SENSOR_VN
EN
VDD
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
UART_TX
UART_RX
JTAG_TDI
JTAG_TCK
JTAG_TMS
JTAG_TDO
EoTLED1
EoTLED2
EoTLED4
EoTLED3
JOY_BTN2
JOY_BTN1
JOY_DOWN
JOY_UP
JOY_X/LEFT
JOY_Y/RIGHT
J5
Button1+2
J4
Down/Up
J3
Left/Right
Microcontroller and supporting components
EoTLED1
EoTLED2
EoTLED3
EoTLED4
J9
EoTSwitchOut
J10
EoTSwitchOut2
J11
EoTSwitchOut3
J12
EoTSwitchOut4
VBAT
Q1
DMP2035U
R6
100K
BT1
Battery
Battery holder and
Reverse Polarity Protection
UART_TX
UART_RX
J1
Conn_01x04
UART header
JTAG_TMS
JTAG_TDI
JTAG_TCK
JTAG_TDO
J2
Conn_01x04
JTAG header
JOY_X/LEFT
JOY_Y/RIGHT
JOY_BTN1
JOY_BTN2
JOY_UP
JOY_DOWN
R8
10K
R7
10K
R11
10K
R12
10K
R10
10K
R9
10K
Pull ups
J6
AnalogStick Pot1 +/−
J7
AnalogStick Pot2 +/−
Optional analog stick potentiometer connections